SC1006 Computer Organisation and Architecture

Parallel Data Transfer

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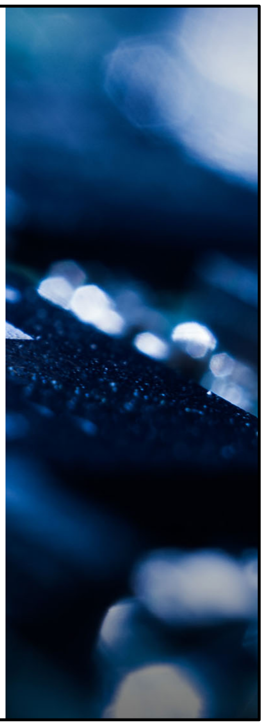
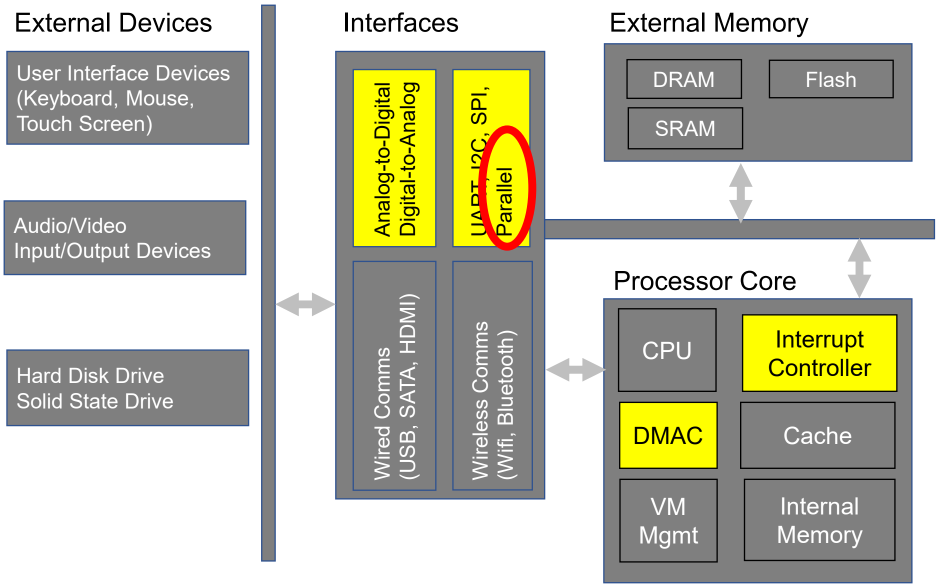


- Background picture
  https://pixabay.com/photos/electrician-electronics-computer-1288717/

- **VO (for approval):**
  This section is about transferring data via the parallel interface.

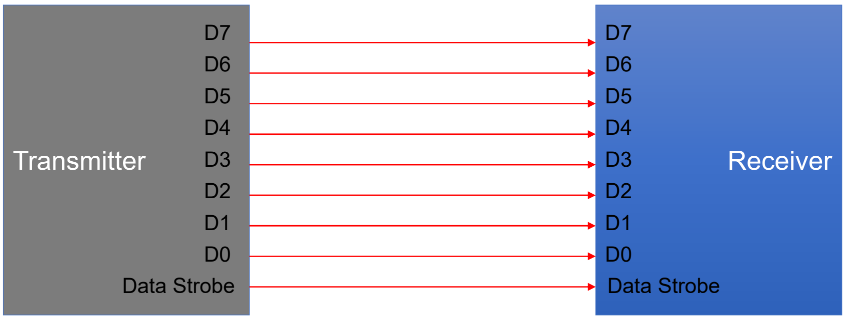
Computer System Block Diagram



VO:
In this video, we will continue, to dive deeper, into these few elements of the computer system.!!

Parallel Data Transfer

- Multiple bits of data are transferred simultaneously between two devices.
- Synchronous in nature as some sort of strobe signal is needed to inform the receiver when to latch in the data. E.g. rising edge of strobe signal.

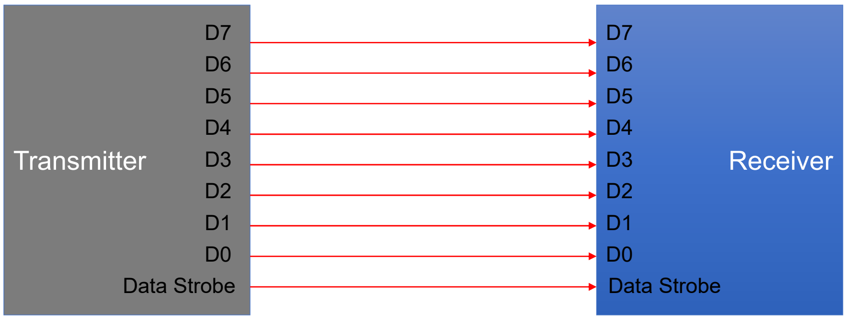


VO:

Data can be transferred, either one bit, or multiple bits, at a time... When multiple bits, are transferred simultaneously, it is known as, a Parallel interface... One example, is shown in this slide.. when the data, is transmitted, on multiple lines collectively, this is known as a data bus... The transfer is typically synchronized, by a strobe signal. as indicated by, the pin Data Strobe, in the diagram...For example, data could be transferred, on the rising edge, of the strobe signal... The strobe signal, may also be, referred to, as the clock signal, of the parallel bus.. when we are comparing, the data transfer rate ,with respect to the serial interface...

Parallel Data Transfer

- Multiple bits of data are transferred simultaneously between two devices.
- Synchronous in nature as some sort of strobe signal is needed to inform the receiver when to latch in the data. E.g. rising edge of strobe signal.
- Able to achieve higher transfer rate than Serial Interface (using same clock).
- But more prone to Signal Skew and Crosstalk (see later slides).

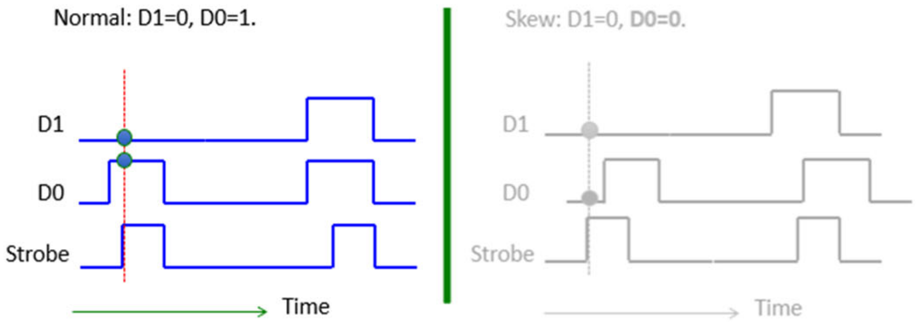


VO:

Some examples, of devices with parallel interface, are Static Random Access Memory, The S.RAM, Dynamic RAM, D.RAM, Read-Only-Memory the ROM etc. Given the same clock rate, Parallel interface, has a faster data rate, than that achievable by the Serial interface...This is because, Parallel interface, has multiple data lines, and therefore, can transfer multiple data bits simultaneously... Serial interface, on the other hand, can only transfer, one bit at a time, since it has only one, data line... However.. because there are more signal lines, that are closed to each other, parallel interface is more Prone , to signal skew, and crosstalk... These two phenomena, will result in wrong data, being latched by the receiver... More details in the next a few slides...

Signal Skew

- If for some reason (due to circuit design, external interference), the signal in one or more data lines took different amount of time to reach the receiver, then there is a skew between the signals in the parallel data bus.
- Below example illustrates the effect of signal skew. Data is latched by the receiver on rising edge of the strobe signal.
- This result in wrong data (D0=0 instead of 1) being latched by the receiver.



- VO:
  This is illustrated, in the diagrams, in this slide... Suppose D0, and D1, forms a 2-bit data bus... If the transmitter output 01,.. i.e.. D1=0, D0=1.

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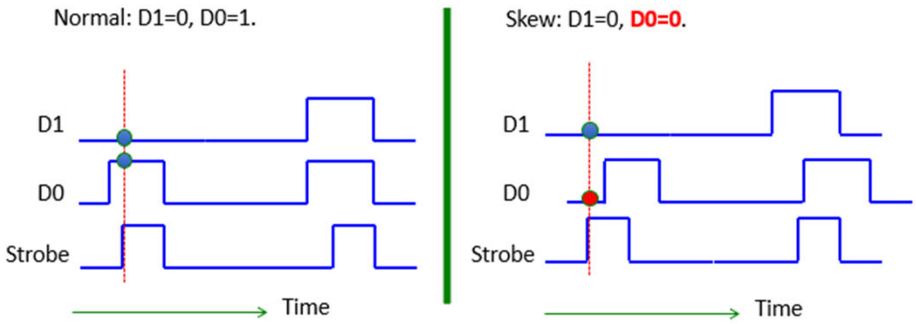
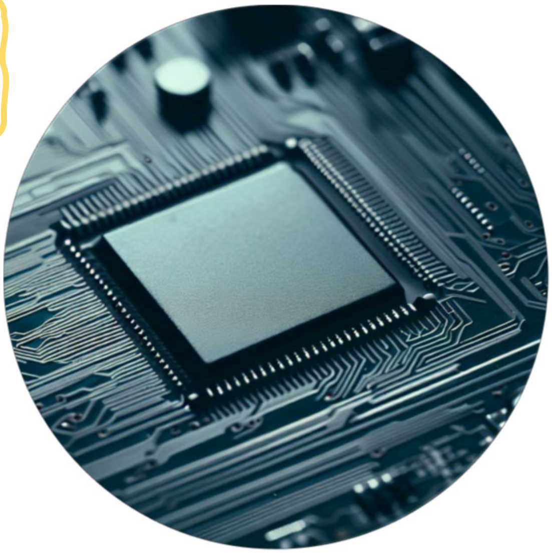


- VO:
  But if, for some reason, which I will elaborate later, the signal on D0, gets delayed,. and reach the receiver, later, compared to D1, and the strobe signal. Then the receiver, will see a zero zero instead..

Signal Skew – Contributing Factors

- Signal Skew is due to variation in propagation delay between signals from the same data bus.
- Propagation delay is the time taken for signal to travel between two points in a circuit.
- Capacitance and Resistance of the physical data line is a major contributor to circuit propagation delay.
  - The larger the resistance and capacitance, the larger the delay. Illustrated in the equation $\tau=RC$ where τ is the time constant dictating rate of change of voltage levels in the data line concerned.
- Variation in resistance and capacitance of the signal lines can be due to
  - Variation in PCB trace length/width (lead to change in impedance).
  - Connecting active components (capacitors, inductors, IC etc) to some of the signal lines.



VO:
So what is propagation delay?

Its the time taken by the electrical signal to travel from one point to another within the electrical circuit.

Many factors can affect the propagation delay.

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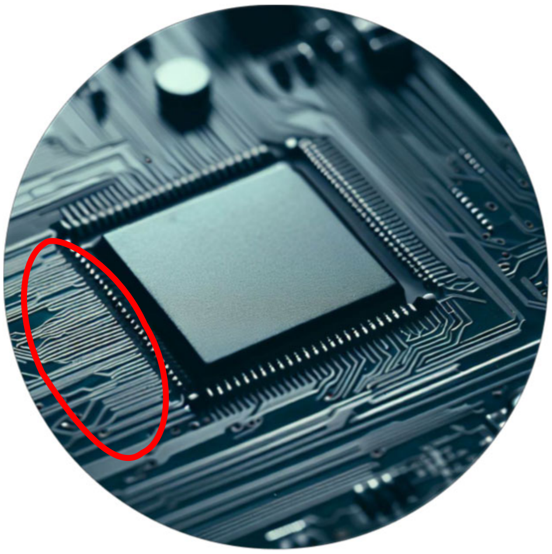

VO:

One of the biggest contributor is capacitance and resistance of the wire that the electrical signal is travelling on.

The product of resistance and capacitance is proportional to the time it takes for any voltage change on the wire to reach its steady state.

What do I mean by that?

Say the logic '1' of the device correspond to 3V, when the device tries to output a logic '1', the signal voltage will not reach 3V immediately but will take some time to get there.

How long it takes is dependent on the RC product value of the circuit it is connected to.

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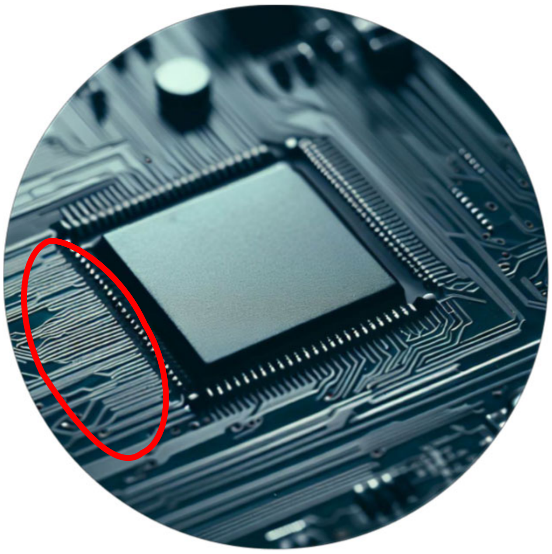
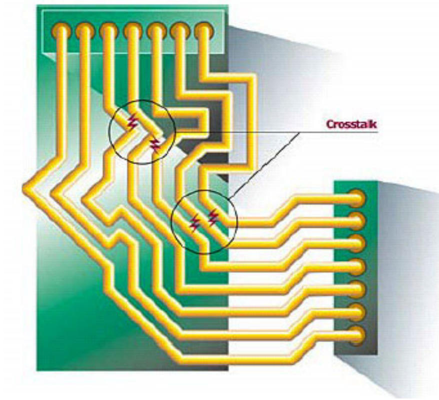


VO:

And these two parameters, change if, the circuit environment changes. Such as variation in length, or width, of PCB traces, Connection of components, such as capacitors, resistors, I.Cs, on the circuit concerned.So when connecting high speed devices, via a parallel bus, it is very important, to ensure that the propagation delay, of the data line, within the same bus, is kept the same, so that data, can be transferred correctly.

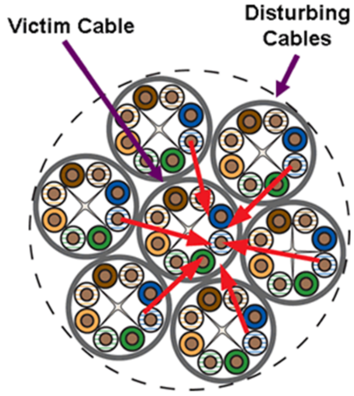
Crosstalk

PCB



Party

Cable



Next is Cross Talk.
In layman's terms, Crosstalk is unwanted interference from neighbors.
In a party where many people are talking simultaneously, voices from people other than the person you are talking to will come across as Crosstalk to you and your partner.

Similarly, in a PCB with many data lines closely packed together, there will be a lot of cross-interference between these signals as they propagate through the circuits.

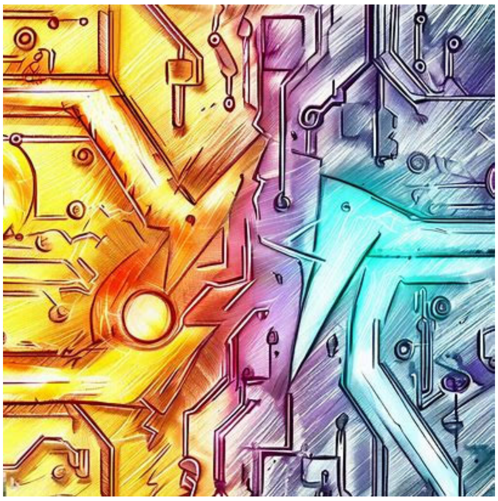
Interference can be via

Electromagnetic domain where the wires behaves like antenna emitting Radio Waves.

Electrical domain as the electrical signal from one wire could potentially coupled to the other wires at the connection point.

They could also be coupled via ground current returning via different paths on the PCB ground plane.

Crosstalk – Electrical Circuit

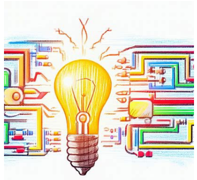


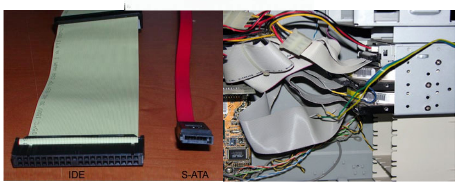
- Crosstalk are undesired coupling of signals from one circuit to another circuit.
- In a parallel bus context, the close placement of the data lines in PCB routing or cabling enables the effect of electrical signal in one trace/wire to be coupled over to the other. Creating undesired interference (crosstalk).
- Crosstalk can be transmitted electrically or via electromagnetic radiation (the trace/wire acts like an antenna).

- This slide summarise what I have talked about in the previous slide.
- You can pause the video for a while to go through the information presented.

Parallel Data Transfer – Pros and Cons

- Advantages
  - Fast data transfer rate (more bits can be transferred at one time)
- Disadvantages
  - Affected by Signal Skew and Cross Talk, which limits the maximum clocking speed and transfer distance.
  - Higher hardware cost compared to Serial data implementation.
    - Hardware (data cable) can be bulky if data width is large.
    - Need more space to route the PCB traces.





- Summarising the pros and cons of parallel interface
- Advantages

- Given the same clock rate, Parallel interface can transfer data at a faster rate compared to Serial Interface.
- However, because there are more signals lines (data and strobe) that are closed to each other, parallel interfaces are more prone to signal skew and crosstalk. These two phenomena will result in wrong data being latched by the receiver. This also limit the maximum clock speed and maximum distance that the signal can travel without being affected by signal skew or cross talk.
- Parallel bus or cable also takes up more space compared to serial bus because more wires are involved.
  - As a result, it is more complex and costly to design the PCB for parallel bus.
- The cable is also bulkier so house keeping can be an issue.